System Analysis

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REQUIREMENTS DETERMINATION

1. What are the key deliverables that are created during analysis?

2. What is the difference between an as-is system and a to-be system?

3. What is the purpose of the requirements definition?

4. What are the three basic steps of the analysis process? Which step is sometimes skipped or done in a cursory fashion? Why?

5. Describe the five major steps in conducting interviews.

6. Explain the differences among a closed-ended question,

an open-ended question, and a probing question.

When would you use each?

7. Explain the differences between unstructured interviews

and structured interviews. When would you use

each approach?

Which is NOT a purpose of the requirements definition?

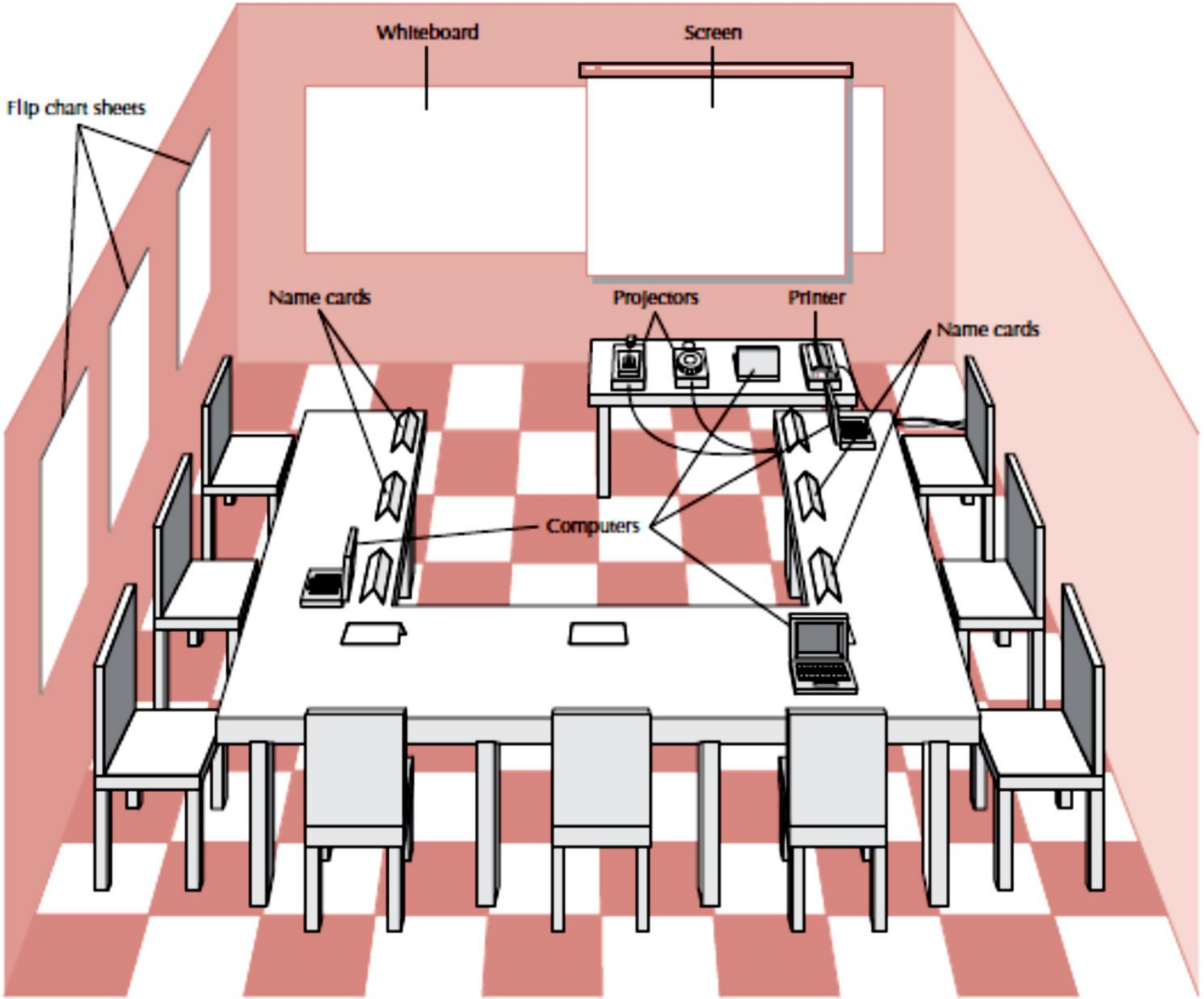
- To give a very high-level explanation of the business requirements
- A more precise list of requirements that can be used as inputs to the rest of analysis
- Create functional requirements
- Create cost/benefit analysis
- Create non-functional requirements

An example of a functional requirement is _____

- Access to the customer order system
- System should be available in English and Spanish
- System can be accessed through a Blackberry device
- Output can be displayed in Internet Explorer, in Firefox, or in Google Chrome browsers
- System is automatically updated every 5 seconds

Joint Application Development (JAD)

JAD is an information-gathering technique that allows the project team, users, and management to work together to identify requirements for the system. IBM developed the JAD technique in the late 1970s, and it is often the most useful method for collecting information from users.


Whiteboard
Screen
Flip chart sheets
Name cards
Projectors
Printer
Name cards
Computers

JAD is a structured process in which ten to twenty users meet together under the direction of a ***facilitator*** skilled in JAD techniques. The facilitator is a person who sets the meeting agenda and guides the discussion but does not join in the discussion as a participant.

He or she does not provide ideas or opinions on the topics under discussion so as to remain neutral during the session. The facilitator must be an expert in both group-process techniques and systems-analysis and design techniques. One or two *scribes* assist the facilitator by recording notes, making copies, and so on. Often the scribes use computers and CASE tools to record information as the JAD session proceedings

A new form of JAD called *electronic JAD*, or *e-JAD*, attempts to overcome these problems by using groupware. In an e-JAD meeting room, each participant uses special software on a networked computer to send anonymous ideas and opinions to everyone else.

In this way, all participants can contribute at the same time without fear of reprisal from people with differing opinions. Initial research suggests that e-JAD can reduce the time required to run JAD sessions by 50 to 80 percent. A good JAD approach follows a set of five steps.

1. Select Participants

First, selecting JAD participants is done in the same basic way as selecting interview participants.

Participants are selected based on the information they can contribute in order to provide a broad mix of organizational levels and to build political support for the new system.

The need for all JAD participants to be away from their office at the same time can be a major problem. The office might need to be closed or operate with a skeleton staff until the JAD sessions are complete.

2. Design a JAD Session

Second, **JAD sessions can run from as little as half a day to several weeks**, depending upon the size and scope of the project. In our experience, most JAD sessions tend to last five to ten days, spread over a three-week period. Most e-JAD sessions tend to last one to four days in a one week period. JAD and e-JAD sessions usually go beyond collecting information and move into analysis. For example, the users and the analysts collectively can create analysis deliverables, such as the functional models or the requirements definition.

As with interviewing, success depends upon a careful plan. JAD sessions usually are designed and structured using the same principles as interviews.

3. Preparing for a JAD Session

Third, as with interviewing, it is important to prepare the analysts and participants for a JAD session. Because the sessions can go beyond the depth of a typical interview and are usually conducted off-site, participants can be more concerned about how to prepare. It is important that the participants understand what is expected of them. If the goal of the JAD session, for example, is to develop an understanding of the current system, then participants can bring procedure manuals and documents with them. If the goal is to identify improvements for a system, then before they come to the JAD session they can think about how they would improve the system.

4. Conducting a JAD Session

Fourth, most JAD sessions **try to follow a formal agenda**, and most have formal *ground rules* that define appropriate behavior. Common ground rules include following the schedule, respecting others' opinions, accepting disagreement, and ensuring that only one person talks at a time.

The role of a JAD facilitator can be challenging. Many participants come to a JAD session with strong feelings about the system to be discussed. Channeling these feelings so that the session moves forward in a positive direction and getting participants to recognize and accept— but not necessarily agree on—opinions and situations different from their own requires significant expertise in systems analysis and design, JAD, and interpersonal skills. Few systems analysts attempt to facilitate JAD sessions without being trained in JAD techniques, and most apprentice with a skilled JAD facilitator before they attempt to lead their first session.

5. Post-JAD Follow-up

Fifth, as with interviews, a JAD *post-session report* is prepared and circulated among session attendees. The post-session report is essentially the same as the interview report. Because the JAD sessions are longer and provide more information, it usually takes a week or two after the JAD session before the report is complete.

Questionnaires

A questionnaire is a set of written questions used to obtain information from individuals.

Questionnaires are often used when there is a large number of people from whom information and opinions are needed. In our experience, questionnaires are a common technique with systems intended for use outside the organization (e.g., by customers or vendors)

1. Select Participants

First, as with interviews and JAD sessions, the first step is to identify the individuals to whom the questionnaire will be sent. However, it is not usual to select every person who could provide useful information. The standard approach is to select a *sample,* or subset, of people who are representative of an entire group. Sampling guidelines are discussed in most statistics books, and most business schools include courses that cover the topic, so we do not discuss it here. The important point in selecting a sample, however, is to realize that not everyone who receives a questionnaire will actually complete it. On average, only 30 to 50 percent of paper and e-mail questionnaires are returned. Response rates for Web-based questionnaires tend to be significantly lower (often only 5 to 30 percent).

2. Designing a Questionnaire

Second, because the information on a questionnaire cannot be immediately clarified for a confused respondent, developing good questions is critical for questionnaires. Questions on questionnaires must be very clearly written and leave little room for misunderstanding, so closed-ended questions tend to be most commonly used. Questions must clearly enable the analyst to separate facts from opinions. Opinion questions often ask respondents the extent to which they agree or disagree (e.g., Are network problems common?), whereas factual questions seek more precise values (e.g., How often does a network problem occur: once an hour, once a day, once a week?). See next Figure for guidelines on questionnaire design. Perhaps the most obvious issue—but one that is sometimes overlooked—is to have a clear understanding of how the information collected from the questionnaire will be analyzed and used. This issue must be addressed before the questionnaire is distributed, because it is too late afterward.

- Begin with nonthreatening and interesting questions.
- Group items into logically coherent sections.
- Do not put important items at the very end of the questionnaire.
- Do not crowd a page with too many items.
- Avoid abbreviations.
- Avoid biased or suggestive items or terms.
- Number questions to avoid confusion.
- Pretest the questionnaire to identify confusing questions.
- Provide anonymity to respondents.

3. Administering the Questionnaire

Third, the key issue in administering the questionnaire is getting participants to complete the questionnaire and send it back. Dozens of marketing research books have been written about ways to improve response rates. Commonly used techniques include clearly explaining why the questionnaire is being conducted and why the respondent has been selected; stating a date by which the questionnaire is to be returned; offering an inducement to complete the questionnaire (e.g., a free pen); and offering to supply a summary of the questionnaire responses. Systems analysts have additional techniques to improve response rates inside the organization, such as personally handing out the questionnaire and personally contacting those who have not returned them after a week or two, as well as requesting the respondents' supervisors to administer the questionnaires in a group meeting.

4. Questionnaire Follow-up

Fourth, it is helpful to process the returned questionnaires and develop a questionnaire report soon after the questionnaire deadline. This ensures that the analysis process proceeds in a timely fashion and that respondents who requested copies of the results receive them promptly.

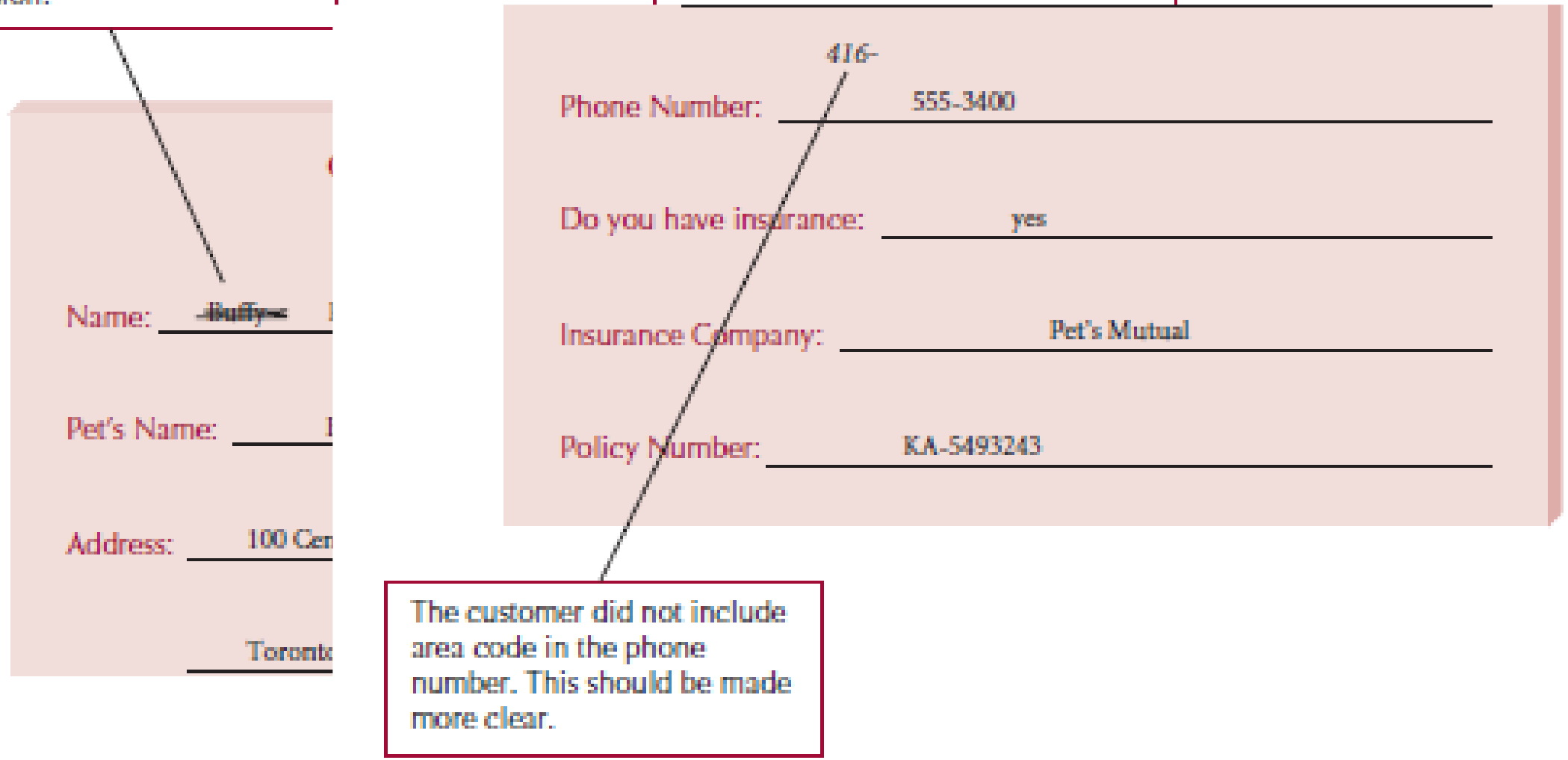
Document Analysis

Project teams often use document analysis to understand the as-is system. Under ideal circumstances, the project team that developed the existing system will have produced documentation that was then updated by all subsequent projects. In this case, the project team can start by reviewing the documentation and examining the system itself.

Unfortunately, most systems are not well documented because project teams fail to document their projects along the way, and when the projects are over, there is no time to go back and document. Therefore, there might not be much technical documentation about the current systems available, or it might not contain updated information about recent system changes. However, many helpful documents do exist in an organization: **paper reports, memorandums, policy manuals, user-training manuals, organization charts, forms, and, of course, the user interface with the existing system.**

The customer made a mistake. This should be labeled **Owner's Name** to prevent confusion.

The staff had to add additional information about the type of animal and the animal's date of birth. This information should be added to the



The customer did not include area code in the phone number. This should be made more clear.

The most powerful indication that the system needs to be changed is when users create their own forms or add additional information to existing ones. Such changes clearly demonstrate the need for improvements to existing systems. Thus, it is useful to review both blank and completed forms to identify these deviations. Likewise, when users access multiple reports to satisfy their information needs, it is a clear sign that new information or new information formats are needed.

Observation/ Ethnography

Observation, the act of watching processes being performed, is a powerful tool for gathering information about the as-is system because it enables the analyst to see the reality of a situation, rather than listening to others describe it in interviews or JAD sessions. Several research studies have shown that many managers really do not remember how they work and how they allocate their time. (Quick, how many hours did you spend last week on each of your courses?) Observation is a good way to check the validity of information gathered from indirect sources such as interviews and questionnaires.

Observation/ Ethnography

Requirements that are derived from the way that people actually work rather than the way I which process definitions suggest that they ought to work.

Requirements that are derived from cooperation and awareness of other people's activities.

- Awareness of what other people are doing leads to changes in the ways in which we do things.

Ethnography is effective for understanding existing processes but cannot identify new features that should be added to a system.

Focused **Observation/ Ethnography**

Developed in a project studying the air traffic control process

Combines ethnography with prototyping

Prototype development results in unanswered questions which focus the ethnographic analysis.

The problem with ethnography is that it studies existing practices which may have some historical basis which is no longer relevant.

Requirements validation

Concerned with demonstrating that the requirements define the system that the customer really wants.

Requirements error costs are high so validation is very important

- Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error.

Requirements checking

Validity. Does the system provide the functions which best support the customer's needs?

Consistency. Are there any requirements conflicts?

Completeness. Are all functions required by the customer included?

Realism. Can the requirements be implemented given available budget and technology

Verifiability. Can the requirements be checked?

Requirements validation techniques

Requirements reviews

- Systematic manual analysis of the requirements.

Prototyping

- Using an executable model of the system to check requirements.

Test-case generation

- Developing tests for requirements to check testability.

Requirements reviews

Regular reviews should be held while the requirements definition is being formulated.

Both client and contractor staff should be involved in reviews.

Reviews may be formal (with completed documents) or informal. Good communications between developers, customers and users can resolve problems at an early stage.

Review checks

Verifiability

- Is the requirement realistically testable?

Comprehensibility

- Is the requirement properly understood?

Traceability

- Is the origin of the requirement clearly stated?

Adaptability

- Can the requirement be changed without a large impact on other requirements?

Requirements management

Requirements management is the process of managing changing requirements during the requirements engineering process and system development.

New requirements emerge as a system is being developed and after it has gone into use.

You need to keep track of individual requirements and maintain links between dependent requirements so that you can assess the impact of requirements changes. You need to establish a formal process for making change proposals and linking these to system requirements.

Changing requirements

The business and technical environment of the system always changes after installation.

- New hardware may be introduced, it may be necessary to interface the system with other systems, business priorities may change (with consequent changes in the system support required), and new legislation and regulations may be introduced that the system must necessarily abide by.

The people who pay for a system and the users of that system are rarely the same people.

- System customers impose requirements because of organizational and budgetary constraints. These may conflict with end-user requirements and, after delivery, new features may have to be added for user support if the system is to meet its goals.

Changing requirements

Large systems usually have a diverse user community, with many users having different requirements and priorities that may be conflicting or contradictory.

- The final system requirements are inevitably a compromise between them and, with experience, it is often discovered that the balance of support given to different users has to be changed.

Requirements evolution

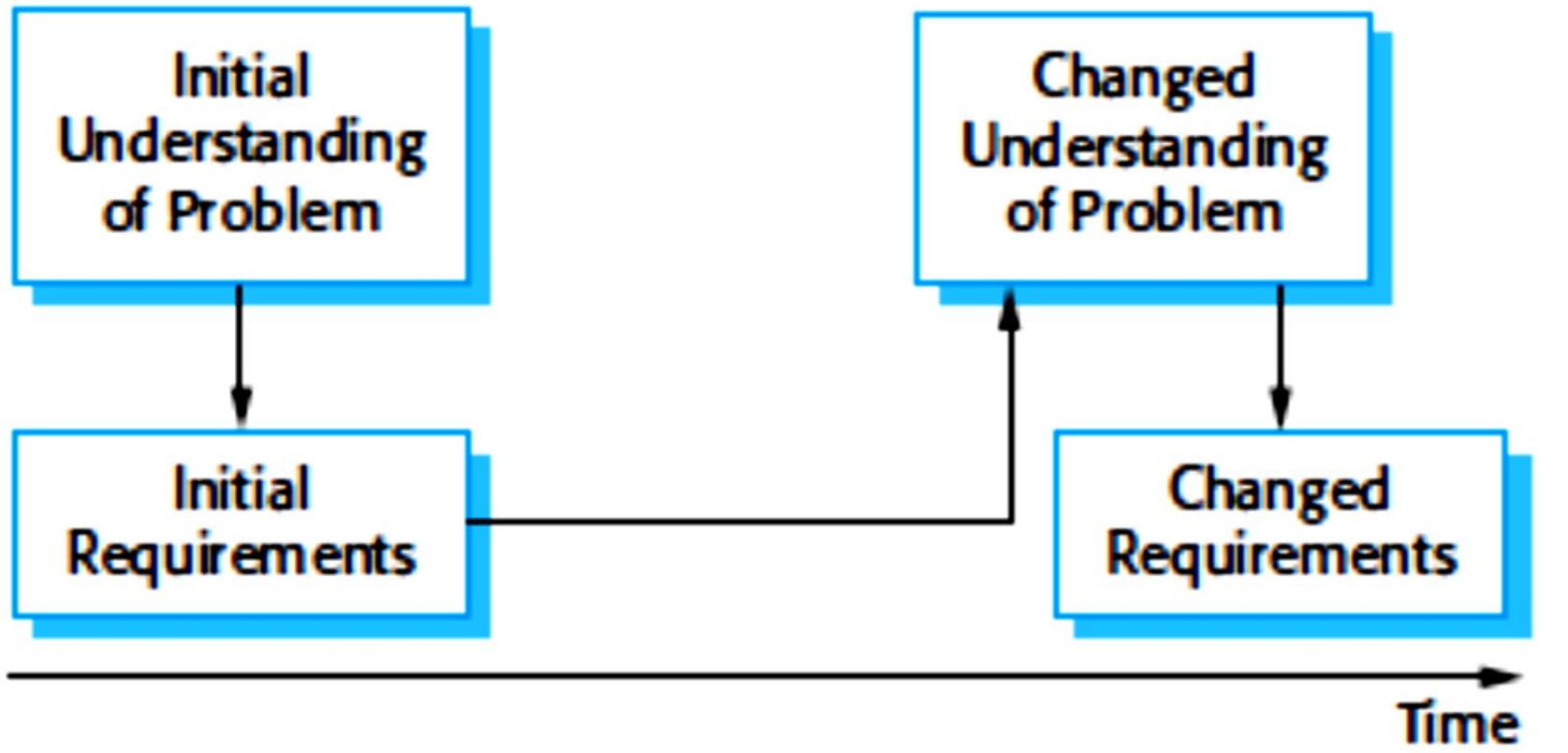


Table of Requirements-Gathering Techniques

	Interviews	Joint Application Design	Questionnaires	Document Analysis	Observation
Type of information	As-is, improvements, to-be	As-is, improvements, to-be	As-is, improvements	As-is	As-is
Depth of information	High	High	Medium	Low	Low
Breadth of information	Low	Medium	High	High	Low
Integration of information	Low	High	Low	Low	Low
User involvement	Medium	High	Low	Low	Low
Cost	Medium	Low–Medium	Low	Low	Low to Medium

System Proposal Template

1. Table of Contents

2. Executive Summary

A summary of all the essential information in the proposal so a busy executive can read it quickly and decide what parts of the proposal to read in more depth.

3. System Request

The revised system request form.

4. Workplan

The original workplan, revised after having completed analysis.

5. Feasibility Analysis

A revised feasibility analysis, using the information from analysis.

6. Requirements Definition

A list of the functional and nonfunctional business requirements for the system

7. Functional Model

An activity diagram, a set of use case descriptions, and a use-case diagram that illustrate the

basic processes or external functionality that the system needs to support.

8. Structural Models

A set of CRC cards, class diagram, and object diagrams that describe the structural aspects of the to-be system. This may also include structural models of the current as-is system that will be replaced.

9. Behavioral Models

A set of sequence diagrams, communication diagrams, behavioral-state machines, and a CRUDE matrix that describe the internal behavior of the to-be system. This may include behavioral models of the as-is system that will be replaced.

10. Appendices

These contain additional material relevant to the proposal, often used to support the recommended system. This might include results of a questionnaire survey or interviews, industry